

VisualLesson AI

Why We Generate, Check, and Compose High-School Lessons in Stages

Technical Product White Paper, Version 2.0

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Abstract

We present VisualLesson AI, a deployable lesson studio that turns a Grades 9–12 question into a structured explanation, source-grounded notes, a quiz, a connected concept map, and an optional narrated video. We ask a practical design question: if generation is probabilistic and media calls are expensive, where should a small educational system check its work? We answer with a staged pipeline. Gate 1 screens the explanation before research and media spending. Gate 2 checks seven image frames with parallel pixel diagnostics and, only when those checks pass, one semantic contact-sheet review. We generate video by composing inspected still images, captions, transitions, and narration rather than by calling a direct text-to-video model. The design follows established ideas from multimedia learning, cognitive-load research, retrieval practice, feedback research, and risk-based AI evaluation, while remaining explicit that automated gates do not prove factual correctness (Mayer, 2009; Sweller, 1988; Roediger and Karpicke, 2006; Hattie and Timperley, 2007; Autio et al., 2024).

This paper documents the complete system as implemented in August 2026. We give accessible explanations in the main text and place the score equations, threshold logic, timing allocation, evaluation metrics, and worked calculations in the appendices. Seven system diagrams show architecture, gate decisions, expected cost, visual sequencing, concept connections, and deployment. We also separate offline checker experiments from live production behavior. The application is available at <https://teachhighschool.streamlit.app/>; source code is available at <https://github.com/allanzhang721/AiLessonStudio>.

1 Introduction

A fluent answer is not automatically a good lesson. It may omit a condition, jump over a reasoning step, use language above the student’s grade, or present a polished but incorrect diagram. These risks matter for high-school learners because a student may not yet have enough background knowledge to notice the error. At the same time, checking everything with another large model can double cost and latency without creating independent evidence.

We built VisualLesson AI around this tension. We do not ask one model call to write an answer, find sources, draw figures, narrate them, and judge the result. We separate these responsibilities. The student first receives an explanation that has passed a local and semantic review. Research notes are generated separately so citations remain visible. The visual pipeline then creates seven inspectable frames. Only after those frames pass Gate 2 do we compose the final MP4.

The uploaded SLaTR manuscript influenced our writing habits: we begin from the uncertainty, define each object before using it, explain the mechanism next to the corresponding diagram, and use “we” when describing choices made by the project team. We do not reproduce its technical claims or wording. This document is a project white paper rather than a peer-reviewed research article.

We make six contributions:

1. We provide a complete public application rather than a notebook-only prototype.
2. We use one provider selection and one visible API-key field. An OpenAI key can serve text, grounded research, images, and narration; DeepSeek is presented honestly as a text-only path in the current product.
3. We implement a local-first Gate 1 that combines transparent structural checks with a compact five-dimension semantic rubric and at most one repair.
4. We implement a local-first Gate 2 that checks seven frames in parallel and makes at most one semantic vision call.
5. We compose video from approved still frames and audio, which makes every teaching step observable and regenerable.
6. We expose evidence, misconceptions, practice, quality records, and related-topic sources instead of returning one uninterrupted block of generated text.

2 Scope, Claims, and Intended Use

We target short explanatory lessons for students in Grades 9–12. The system can support self-study, teacher preparation, and exploration of a difficult concept. It is not a teacher, textbook, examination authority, or diagnostic instrument. A passing gate means that the artifact met the implemented checks; it does not mean that the artifact is certainly correct.

Table 1 separates claims we can support from claims we do not make.

Supported project claim	Claim outside the evidence
The code runs two staged quality gates and records their decisions.	A passing gate guarantees scientific truth or classroom safety.
The saved experiments compare named methods on their recorded datasets.	Offline accuracy equals real-world lesson correctness.
The image-composition path exposes seven inspectable frames.	Seven frames are optimal for every subject or learner.
The quiz provides immediate item feedback and review suggestions.	One five-item quiz estimates long-term mastery.
Visitor keys are not intentionally written to project files or manifests.	The application controls a third-party provider after a request is sent.

Table 1: The paper treats quality gates as risk-reduction controls, not proofs.

3 Evidence Behind the Learning Design

3.1 Why we divide a lesson into visible parts

Cognitive-load research distinguishes useful reasoning from processing consumed by the presentation itself. Sweller’s early analysis argues that demanding problem-solving procedures can consume capacity that would otherwise support schema acquisition (Sweller, 1988). Multimedia-learning research likewise reports benefits from removing irrelevant material, signaling structure, and coordinating words with graphics (Mayer, 2009; Fiorella and Mayer, 2021). We translate these findings into conservative product choices: a focused explanation, explicit key ideas, a worked example, a reserved caption band, a limited number of visual steps, and restrained motion.

These sources do not prove that our interface improves learning. They explain why the interface is structured for inspection and low extraneous load. A classroom evaluation is still required.

3.2 Why we include retrieval and feedback

Retrieval practice can improve delayed retention compared with repeated study, although the effect depends on the task and context (Roediger and Karpicke, 2006). Feedback can be powerful, but its effect depends on what it says and how it is used (Hattie and Timperley, 2007). We therefore include five short questions, answer explanations, and a list of missed concepts. We do not turn one quiz into an opaque mastery score.

3.3 Why we treat generated content as managed risk

Large language models create opportunities for education but also introduce factual, pedagogical, privacy, and overreliance risks (Kasneci et al., 2023). NIST frames generative-AI assurance as an ongoing process of mapping, measuring, and managing risk rather than as a one-time certificate (Autio et al., 2024). Our gates follow the same practical direction: define narrow checks, record what ran, block expensive downstream work when an upstream artifact fails, and preserve a human role.

4 System Overview

Let a request be

$$x = (q, s, g, \ell, p, m, k), \quad (1)$$

where q is the question, s the subject, g the grade, ℓ the language, p the provider, m the selected model, and k the session-only API key. The output is a lesson bundle

$$y = (e, r, z, G, F, a, v, h), \quad (2)$$

where e is the approved explanation, r the research report, z the quiz, G the concept graph, $F = (I_1, \dots, I_7)$ the frame sequence, a the narration, v the MP4, and h the quality history.

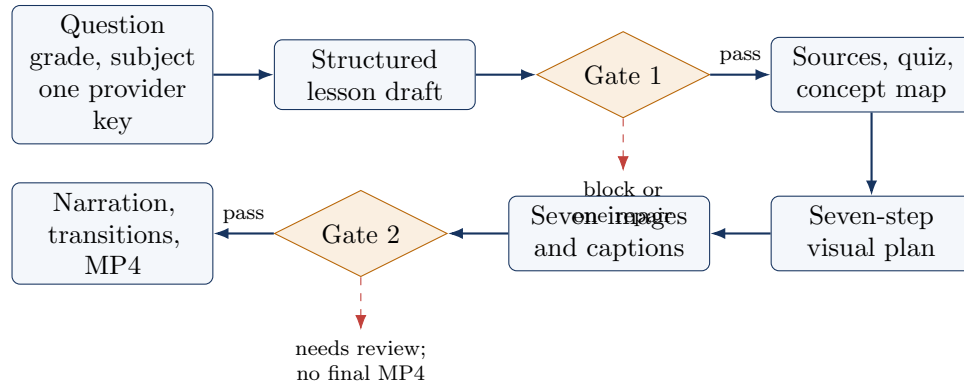


Figure 1: Production architecture. The ordering is also a spending policy: text is checked before image calls, and frames are checked before final composition.

The application stores explicit status rather than a single success flag. Gate 2 can be waiting, blocked by Gate 1, not applicable to a text-only lesson, incomplete because media failed, passed, or marked for review. This prevents a narration failure from erasing a completed visual audit.

5 Why We Use a Staged Cascade

Suppose a text check costs C_T , research costs C_R , images cost C_I , and narration plus composition costs C_V . Let p_1 be the fraction of drafts that pass Gate 1 and p_2 the fraction of frame sets that pass Gate 2.

A simplified expected variable cost is

$$\mathbb{E}[C] = C_T + p_1(C_R + C_I) + p_1p_2C_V. \quad (3)$$

The equation uses multiplication for a simple reason: the video stage is reached only when both earlier events occur. If a draft fails, we do not buy images for it. If its images fail, we do not pretend the final video is approved.

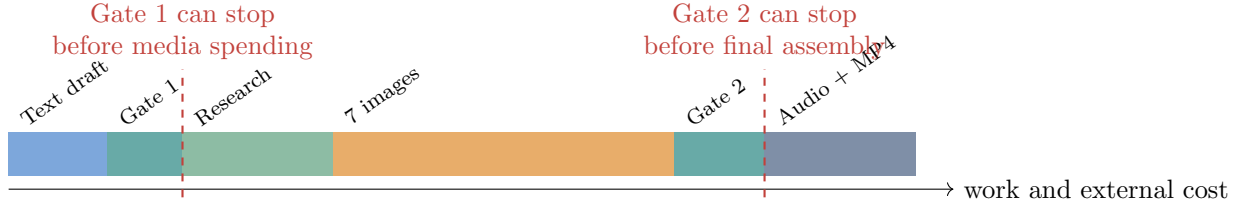


Figure 2: Qualitative cost path. Segment widths illustrate relative workflow weight; they are not provider prices. Actual prices and latency vary by model and date.

The cascade also improves diagnosis. A local readability warning, a factual rubric failure, a dark frame, and a sequence-continuity problem should not all appear as “generation failed.” Each belongs to a different stage and suggests a different correction.

6 Gate 1: Explanation Review

6.1 Local screen

The local screen checks four transparent conditions: the explanation contains 70–420 alphabetic words; it includes at least one important term from the question when such terms exist; and its mean sentence length is no more than 27 words. These values are engineering defaults for a focused lesson, not universal reading-level standards. The gate reports the violated condition instead of hiding it behind a classifier label.

6.2 Semantic rubric and bounded repair

A semantic reviewer scores accuracy, completeness, logical flow, grade fit, and clarity from 1 to 4. Accuracy must equal 4, and every other dimension must be at least 3. When the first review fails but returns a revision, the system allows one repair and one recheck. The maximum of two semantic calls bounds the loop.

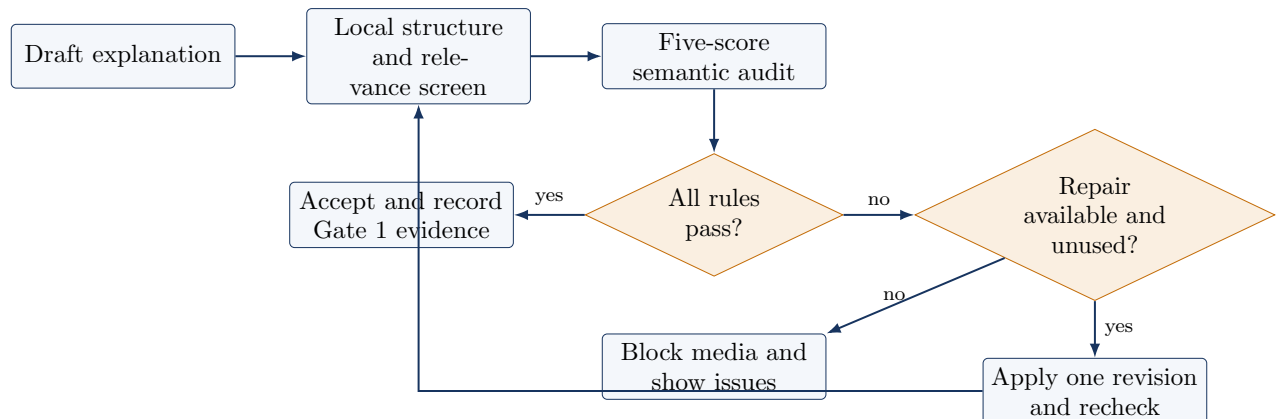


Figure 3: Gate 1 decision graph. The loop is bounded to prevent indefinite self-revision.

If the five scores are $s_d \in \{1, 2, 3, 4\}$, the displayed normalized score is

$$S_1 = \frac{1}{5} \sum_{d=1}^5 \frac{s_d}{4}. \quad (4)$$

A score of $S_1 = 0.90$ is not sufficient by itself. The logical pass rule still requires maximum accuracy and no failed local check. Appendix A gives the exact indicator form and examples.

6.3 Why the saved text classifier is not the production judge

The repository retains DistilBERT, BERT, and RoBERTa experiments. BERT introduced bidirectional transformer pretraining, RoBERTa revised its training recipe, and DistilBERT compressed a larger teacher model (Devlin et al., 2019; Liu et al., 2019; Sanh et al., 2019). Our saved models classify pedagogical error categories on the experiment dataset. That is useful offline evidence, but it is not the same task as verifying an arbitrary new explanation. We therefore report those models as baselines and use the narrower, inspectable cascade in production.

7 Gate 2: Visual Review

7.1 Parallel technical checks

Gate 2 receives seven ordered frames. For each frame, it measures width, height, mean grayscale brightness, grayscale contrast, edge variation, and variation in the lower caption band. Four worker threads process frames in parallel. Five normalized component scores then represent resolution, exposure, contrast, detail, and caption-band readability. Their weighted sum produces a technical frame score $t_i \in [0, 1]$.

The local stage uses a strict all-frame rule. Its mean must be adequate, and every required frame must meet the technical threshold 0.42. This is important because an average can hide one unusable teaching step. If any frame fails, the system stops before a vision-model call.

7.2 One semantic contact-sheet review

When the technical stage passes and a compatible vision client is available, the frames are resized into one ordered contact sheet. One call scores semantic accuracy, teaching alignment, sequence continuity, legibility, and visual load from 1 to 4. Semantic accuracy must equal 4 and all other dimensions must be at least 3. The combined display score is

$$S_2 = 0.35T + 0.65M, \quad (5)$$

where T is mean technical quality and M is normalized semantic quality. The larger semantic weight expresses an engineering priority: a sharp, bright image is still a poor teaching image if it communicates the wrong idea. The weights and thresholds were not learned from classroom outcomes and should be recalibrated when a suitable labeled dataset becomes available.

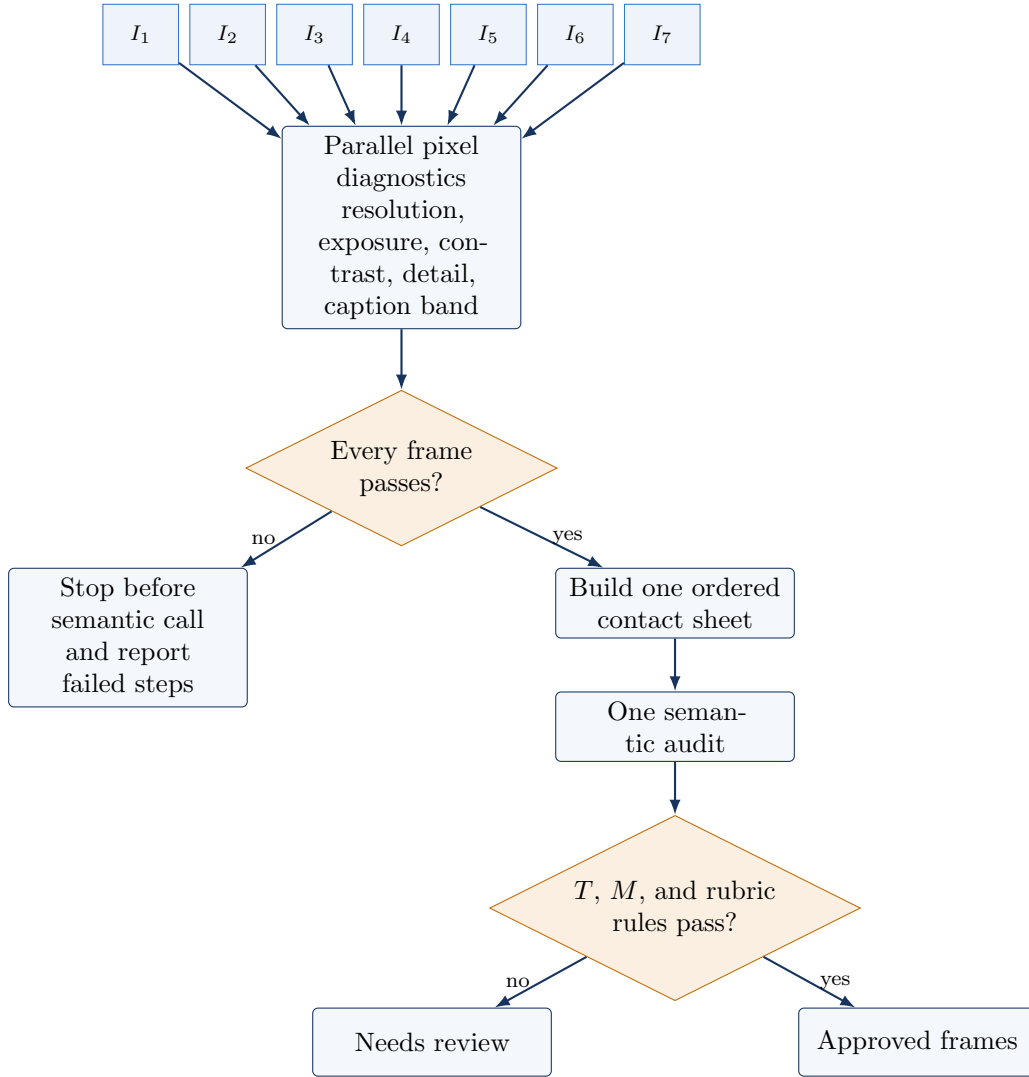


Figure 4: Gate 2 decision graph. Pixel diagnostics screen rendering failures; the contact sheet addresses meaning and continuity.

7.3 Why CLIP remains an offline comparison

CLIP learns an image–text representation through contrastive pretraining and can measure whether an image and caption are close in its embedding space (Radford et al., 2021). Our saved Gate 2 experiments compare cosine thresholding with linear and multilayer probes on aligned and mismatched Flickr8k pairs (Hodosh et al., 2013). This benchmark asks whether a caption matches an image. Production Gate 2 also asks whether labels are readable, exposure is usable, the sequence remains coherent, and the visual teaches the approved explanation. Because the tasks differ, we do not present the CLIP benchmark as a production certification.

8 From Approved Frames to Video

8.1 Why we removed direct text-to-video generation

We removed the direct video path for three reasons. First, a generated clip is harder to inspect at the level of individual teaching steps. Second, it can change objects, labels, or causal relations over time.

Third, its cost is difficult to control in a classroom workflow. The still-image route creates seven explicit checkpoints and allows visual-only regeneration without rewriting an approved explanation.

8.2 Storyboard contract

The planner produces a shared scene bible and seven ordered steps. Each step states its teaching goal, retained objects, additions, removals, labels, and misconceptions to avoid. The image prompt includes the same visual style, layout zones, typography, color assignments, and canonical explanation. A fixed lower band is reserved for captions. These constraints implement coherence and signaling as engineering rules rather than asking the image model to rediscover them for every frame (Fiorella and Mayer, 2021).

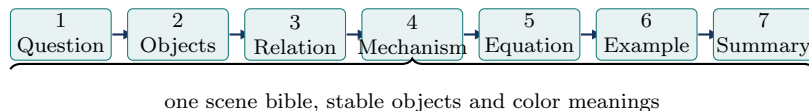


Figure 5: A seven-scene instructional timeline. Labels describe roles, not a mandatory subject template; the planner adapts them to the question.

8.3 Narration-aware timing and composition

The system estimates or measures narration length, gives every scene at least 3.2 seconds, and allocates remaining time by caption word count. Frames are fitted to a 16:9 canvas over a blurred background so content is not cropped. A restrained 1.8% Ken Burns movement alternates direction, and adjacent scenes cross-fade. The video is encoded as H.264 with CRF 18, browser-compatible yuv420p, and fast-start metadata. Narration is normalized toward -16 LUFS with a -1.5 dB true-peak ceiling when the FFmpeg filter is available. Appendix A gives the timing equation.

9 Student-Facing Lesson Structure

The generated bundle is rendered as six views:

1. **Lesson:** objective, explanation, key ideas, worked example, common mistake, and quick check.
2. **Concept map:** prerequisites, applications, contrasts, and next topics.
3. **Evidence and sources:** a grounded synthesis, source-by-source notes, and evidence limits.
4. **Practice:** five multiple-choice items with answer explanations and missed-concept guidance.
5. **Learning library:** deterministic links to established platforms rather than model-invented URLs.
6. **Quality report:** gate path, score, latency, issues, repairs, and model-call count.

Markdown and common LaTeX wrappers are normalized before Streamlit renders them. This prevents a useful equation such as $F = ma$ from appearing as raw control characters. The normalizer preserves paragraphs and lists instead of flattening the lesson into one line.

9.1 Connected concepts with bounded links

The text model proposes five short topic connections and classifies each as a prerequisite, application, contrast, or next step. The program discards model-generated URLs. It attaches sources using deterministic subject mappings to platforms such as OpenStax, Khan Academy, GeoGebra, PhET, and MDN.

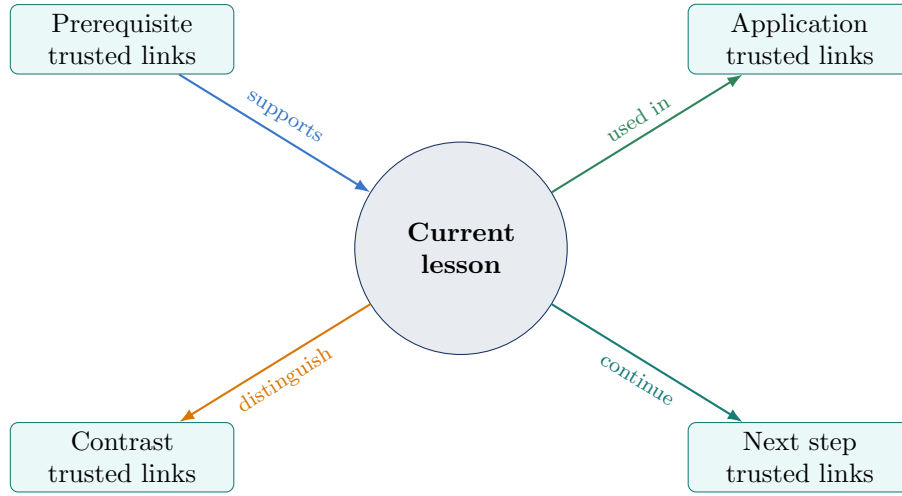


Figure 6: Concept neighborhood. The model proposes instructional relationships; code-controlled mappings determine destination domains.

10 Offline Checker Experiments

The repository contains saved benchmark summaries for the earlier research checkers. Table 2 reports the values used by the application’s friendly experiment snapshot. Gate 1 and Gate 2 use different datasets and tasks, so their rows must not be ranked against one another.

Gate	Method	Accuracy	Precision	Recall	F1	AUROC
1	DistilBERT classifier	0.9280	0.9444	0.9280	0.9284	–
1	BERT classifier	0.9253	0.9314	0.9253	0.9249	–
1	RoBERTa classifier	0.9653	0.9667	0.9653	0.9655	–
2	CLIP cosine threshold	0.9628	0.9474	0.9800	0.9634	0.9922
2	CLIP linear probe	0.8756	0.8301	0.9444	0.8836	0.9555
2	CLIP MLP probe	0.9517	0.9462	0.9578	0.9520	0.9885

Table 2: Saved offline metrics. Gate 1 classifies pedagogical error types. Gate 2 classifies aligned versus mismatched Flickr8k pairs. “–” means AUROC was not retained in the summarized multiclass result.

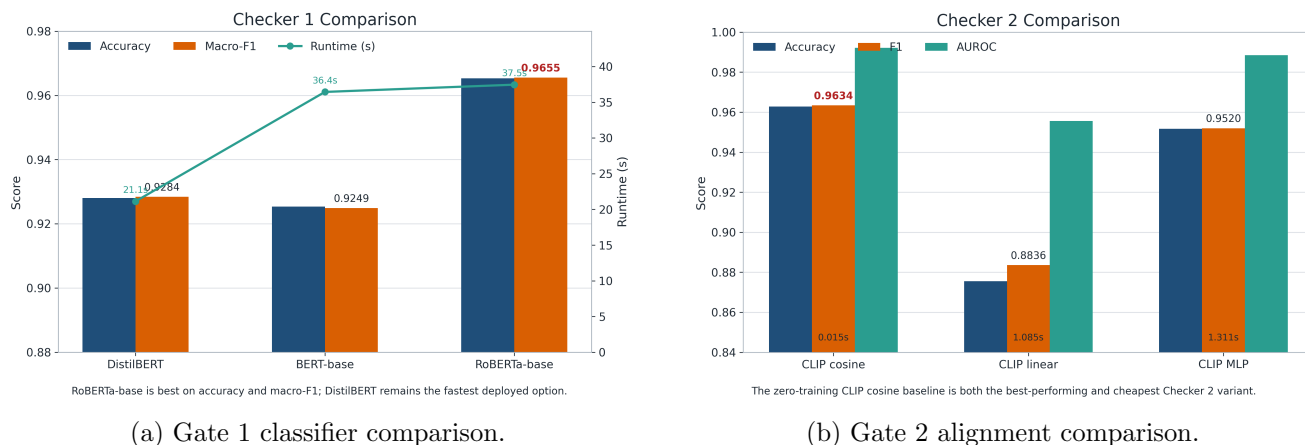


Figure 7: Offline experiment figures generated from repository records. These plots do not show live request latency or learning outcomes.

The RoBERTa classifier has the highest saved Gate 1 F1. The untrained CLIP cosine threshold has the highest saved Gate 2 F1 and AUROC. This result illustrates why “trained” is not itself a deployment metric. We also consider task match, startup weight, latency, interpretability, and whether a baseline measures the product failure we need to catch.

10.1 Metric interpretation

Accuracy asks what fraction of all predictions were correct. Precision asks, among examples predicted positive, how many were positive. Recall asks, among all positive examples, how many were found. F1 is the harmonic mean of precision and recall. AUROC summarizes ranking across thresholds. Appendix A defines each quantity and gives a small confusion-matrix example. For imbalanced classes, accuracy alone can be misleading; this is why we retain multiple metrics (Fawcett, 2006).

11 Reproducible Local Setup

11.1 Requirements

The production target is Python 3.12. The minimal application dependencies are OpenAI’s Python SDK, Pillow, ImageIO, ImageIO-FFmpeg, Streamlit, Requests, and Graphviz. A local Graphviz executable may also be required for graph rendering. Research and training dependencies are isolated from the website requirements.

11.2 Installation

```
git clone https://github.com/allanzhang721/AiLessonStudio.git
cd AiLessonStudio
python -m venv .venv
# Windows PowerShell
.venv\Scripts\Activate.ps1
# macOS or Linux: source .venv/bin/activate
python -m pip install --upgrade pip
python -m pip install -r requirements.txt
python -m pytest -q
python -m streamlit run streamlit_app.py
```

The app opens even before a key is entered, but generation remains locked. The sidebar shows one provider selector and one password-style key field. With OpenAI, the same key is reused for supported text, grounded research, image, and narration calls. With DeepSeek, the app exposes text generation and disables illustrated video because the current pipeline has no same-provider DeepSeek image and narration path. We do not ask for a second provider credential.

11.3 Directory map

```
streamlit_app.py # production entrypoint
app_v2.py # complete Streamlit interface
pipeline/
  lesson_service.py # bundle generation, research, sources
  quality_gates.py # Gate 1
  frame_checker.py # Gate 2
  planner.py # seven-scene plan
  image_pipeline.py # frame generation
  video_pipeline.py # timing, motion, audio, MP4
  markdown_render.py # Markdown and LaTeX normalization
  gate_benchmarks.py # small offline metric snapshot
tests/ # deployment and regression tests
requirements.txt # production dependencies
requirements-research.txt # optional experiment dependencies
```

11.4 Test protocol

The deployment suite contains 31 tests at this paper version. It covers frame scoring and short-circuit behavior, explanation repair, provider routing, one-key Streamlit rendering, source annotation conversion, schema normalization, Markdown and LaTeX handling, fail-fast plan validation, student-analysis helpers, narration arguments, audio-aware timing, and real H.264 encoding. The command used for the deployment branch is:

```
python -m pytest \
  tests/test_frame_checker.py tests/test_production_quality.py \
  tests/test_provider_selection.py tests/test_render_failfast.py \
  tests/test_video_quality.py tests/test_validation.py \
  tests/test_student_analyzer.py -q
```

A passing unit suite establishes that the code meets the tested contracts. It does not establish educational effectiveness. A publication evaluation should additionally include expert factual review, student comprehension measures, accessibility review, cost records, and failure analysis.

12 Deployment, Privacy, and Operational Boundaries

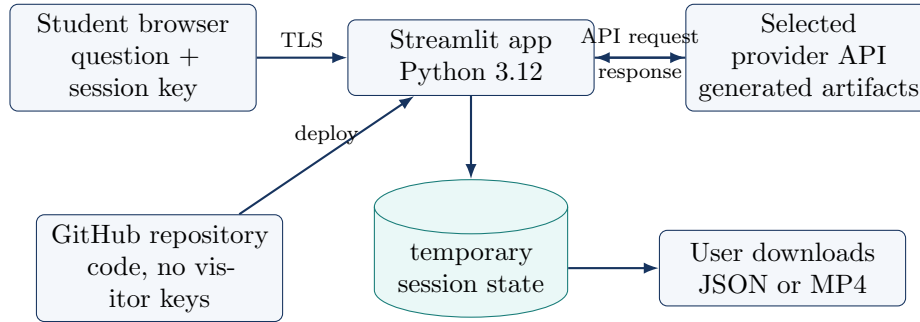


Figure 8: Deployment and data path. The project does not intentionally persist visitor keys; external providers still receive the requests authorized by those keys.

The public deployment uses the [AiLessonStudio repository](#). Its production coordinates are branch `agent/streamlit-production`, entrypoint `streamlit_app.py`, and Python 3.12. Streamlit Community Cloud installs dependencies from the repository and supports configuration of Python and secrets through its deployment settings (Streamlit, 2026a,b). The current visitor flow does not require a repository secret because the visitor provides one session key.

The password-style field reduces accidental display but is not a complete security boundary. The key is passed to the chosen provider client. Students should use restricted keys, set provider-side budgets, avoid entering personal information in prompts, and revoke a key they believe was exposed. School deployments should perform their own privacy, consent, procurement, and retention review.

13 Source and Figure Provenance

Table 3 records where each kind of paper evidence comes from. This is important because a system diagram, an experiment plot, and an external research claim have different evidentiary status.

Artifact	Source	Interpretation
Figures 1–8	Drawn in TikZ from the production source code and deployment configuration	Explanatory system diagrams; not experimental observations
Figure 7	Repository plotting outputs and saved benchmark values	Offline method comparison on recorded tasks
Table 2	<code>pipeline/gate_benchmarks.py</code>	Distilled values used by the public dashboard
Gate equations	<code>quality_gates.py</code> , <code>frame_checker.py</code> , and <code>video_pipeline.py</code>	Exact implementation as of the paper version
Learning rationale	Cited books, peer-reviewed articles, and public standards	Motivation for design choices, not a direct evaluation of this app
Setup instructions	Repository files and official Streamlit documentation	Reproducibility and deployment procedure

Table 3: Evidence and figure provenance. No generated stock artwork is used in the technical diagrams.

14 Limitations and Threats to Validity

Shared-model bias. A generator and semantic reviewer from related model families can share mistakes. The second call is a structured audit, not independent expert replication.

Proxy measurements. Brightness, contrast, and edge variation detect technical defects but do not establish scientific correctness. Likewise, an image-caption benchmark does not measure the learning value of a seven-frame lesson.

Threshold selection. The production thresholds are transparent engineering defaults. They have not been calibrated on a representative sample of teacher-rated generated lessons. Appendix C lists each threshold so it can be challenged and replaced.

Dataset mismatch. Gate 1 offline models classify error categories. Gate 2 offline models use Flickr8k alignment. Neither dataset estimates the probability that a new lesson improves high-school comprehension.

Provider dependence. Model behavior, availability, price, and policy can change. The paper records product logic rather than promising a permanent provider capability.

Accessibility and language. The interface supports multiple output languages, but the structural word rules are based on English alphabetic tokens. Caption readability metrics do not replace screen-reader, color-contrast, hearing-access, or multilingual review.

No classroom trial. We have not yet reported a randomized or controlled student study. The correct next publication step is an evaluation with teachers and learners, not stronger language about the current benchmarks.

15 Recommended Publication Evaluation

We recommend a preregistered evaluation with three layers.

1. **Artifact review:** subject teachers rate factual accuracy, missing conditions, grade appropriateness, and diagram correctness for a stratified set of questions.
2. **System review:** measure latency, provider cost, gate pass rate, repair rate, false blocks, missed errors, and failure recovery.
3. **Learner study:** compare an approved text lesson, an approved text-plus-video lesson, and a text-book/control condition using immediate and delayed questions. Record prior knowledge and avoid claiming personalization from one quiz.

For gate calibration, teachers should label individual failure types and final usability independently. We can then choose thresholds on a validation split and report confidence intervals on a held-out test split. Precision and recall should be reported for the harmful-artifact class because missing a bad lesson and unnecessarily blocking a good lesson have different costs.

16 Conclusion

We built VisualLesson AI as a staged lesson studio because generated educational artifacts fail in different ways and because later media stages are costly. Gate 1 checks the explanation before research and images. Gate 2 checks every frame before final composition. The video path keeps seven inspectable images, stable

captions, controlled timing, and a recoverable intermediate state. The app adds sources, misconceptions, practice, related concepts, regeneration, and a visible quality report around the generated explanation.

The central lesson is not that more checking guarantees truth. It is that narrow checks, placed at the right boundaries, make failures cheaper to detect and easier to explain. The offline classifiers remain useful comparisons. The live gates remain engineering controls. Teacher and student evaluation remains necessary. Keeping these statements separate makes the project easier to reproduce, criticize, and improve.

A Mathematical Appendix: Gate and Timing Equations

This appendix begins with the required high-school mathematics and then states the implementation exactly.

A.1 A short primer

A *weighted average* is like a course grade in which categories have different importance. If homework counts for 30% and an exam counts for 70%, the grade is $0.30H + 0.70E$. Our visual score uses the same idea. An *indicator* $\mathbf{1}[A]$ equals 1 when statement A is true and 0 otherwise. Indicators let us write several pass rules in one equation. A *threshold* is a boundary: a score passes when it is at least the threshold. Thresholds trade false acceptance against false rejection.

A.2 Gate 1

Let the five rubric scores be

$$\mathbf{s} = (s_a, s_c, s_l, s_g, s_r), \quad s_d \in \{1, 2, 3, 4\},$$

for accuracy, completeness, logical flow, grade fit, and clarity/readability. The displayed score is

$$S_1 = \frac{s_a + s_c + s_l + s_g + s_r}{20}. \quad (6)$$

Let J be the number of local issues. The exact pass indicator is

$$P_1 = \mathbf{1}[J = 0] \mathbf{1}[s_a = 4] \mathbf{1}[\min(s_c, s_l, s_g, s_r) \geq 3] \mathbf{1}[b = 1], \quad (7)$$

where b is the semantic reviewer's explicit Boolean decision. Requiring both numeric rules and $b = 1$ guards against an internally inconsistent response.

Worked example. Suppose the scores are $(4, 3, 3, 4, 3)$. Then

$$S_1 = \frac{4 + 3 + 3 + 4 + 3}{20} = \frac{17}{20} = 0.85.$$

If $J = 0$ and $b = 1$, the explanation passes. Now suppose the scores are $(3, 4, 4, 4, 4)$. The average is $19/20 = 0.95$, but the explanation fails because accuracy is not 4. This shows why an average alone is not the decision.

The maximum semantic call count is

$$N_{\text{calls},1} \leq 1 + R_{\text{max}} = 2, \quad (8)$$

with $R_{\text{max}} = 1$. A repair occurs only after a failed first review that supplies a nonempty revision.

A.3 Gate 2 technical score

Define the clipping function

$$\text{clip}_{[0,1]}(x) = \max(0, \min(1, x)) \quad (9)$$

and the range score

$$\rho(x; a, b) = \text{clip}_{[0,1]} \left(\frac{x - a}{b - a} \right), \quad b > a. \quad (10)$$

For frame i , let w_i, h_i be dimensions, μ_i mean grayscale brightness, σ_i grayscale standard deviation, e_i edge-map standard deviation, and c_i caption-band standard deviation. The five component scores are

$$r_i = \begin{cases} 1, & w_i \geq 1200 \text{ and } h_i \geq 800, \\ 0.4, & \text{otherwise,} \end{cases} \quad (11)$$

$$x_i = \min\{\rho(\mu_i; 18, 70), \rho(255 - \mu_i; 18, 70)\}, \quad (12)$$

$$q_i = \rho(\sigma_i; 18, 60), \quad (13)$$

$$d_i = \rho(e_i; 14, 46), \quad (14)$$

$$b_i = \rho(c_i; 8, 28). \quad (15)$$

The frame score is

$$t_i = 0.20r_i + 0.20x_i + 0.25q_i + 0.25d_i + 0.10b_i. \quad (16)$$

The weights sum to 1, so $t_i \in [0, 1]$. For $n = 7$ frames,

$$T = \frac{1}{n} \sum_{i=1}^n t_i, \quad P_T = \mathbf{1}[T \geq 0.42] \prod_{i=1}^n \mathbf{1}[t_i \geq 0.42]. \quad (17)$$

The product equals 1 only when every frame passes.

Worked example. Suppose one frame has component scores (1.00, 0.80, 0.70, 0.60, 0.90). Then

$$t_i = 0.20(1) + 0.20(0.8) + 0.25(0.7) + 0.25(0.6) + 0.10(0.9) = 0.775.$$

It passes the technical threshold. If the seven-frame mean is 0.78 but one other frame scores 0.39, the set fails because the all-frame rule catches the weak step.

A.4 Gate 2 semantic and combined score

Let the five semantic rubric scores be $v_k \in \{1, 2, 3, 4\}$. Then

$$M = \frac{1}{20} \sum_{k=1}^5 v_k, \quad S_2 = 0.35T + 0.65M. \quad (18)$$

If v_a denotes semantic accuracy, the final pass rule is

$$P_2 = P_T \mathbf{1}[v_a = 4] \mathbf{1} \left[\min_k v_k \geq 3 \right] \mathbf{1}[b_2 = 1] \mathbf{1}[S_2 \geq 0.65]. \quad (19)$$

For example, $T = 0.78$ and rubric scores (4, 3, 3, 4, 3) give $M = 0.85$ and

$$S_2 = 0.35(0.78) + 0.65(0.85) = 0.8255.$$

The set passes if the Boolean and all-frame rules also pass. As in Gate 1, the combined number cannot override a required dimension.

A.5 Expected cost of the cascade

Equation (3) assumes that later stages run conditionally. With illustrative units $C_T = 1$, $C_R = 1$, $C_I = 7$, $C_V = 2$, $p_1 = 0.8$, and $p_2 = 0.9$,

$$\mathbb{E}[C] = 1 + 0.8(1 + 7) + (0.8)(0.9)(2) = 8.84.$$

Always running every stage would cost $1 + 1 + 7 + 2 = 11$ units. The difference, 2.16 units in this toy example, comes from not processing blocked artifacts. These are illustrative units, not provider prices.

A.6 Narration-aware scene duration

Let n be the number of captions and

$$w_i = \max(5, \text{caption word count}_i). \quad (20)$$

If measured narration duration A is available, define

$$D = \max(A + 0.65, 3.2n). \quad (21)$$

Otherwise the code estimates

$$A = \max\left(3.2n, \frac{\sum_i w_i}{2.35} + 1.2\right) \quad (22)$$

and applies the same D rule. Starting from a 3.2-second floor, the duration of scene i is

$$d_i = 3.2 + \max(0, D - 3.2n) \frac{w_i}{\sum_j w_j}. \quad (23)$$

Thus every scene receives reading time, while wordier captions receive a larger share of the remaining duration.

B Evaluation Metrics and a Worked Confusion Matrix

For binary labels, let TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives. Then

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad (24)$$

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (25)$$

$$\text{Recall} = \frac{TP}{TP + FN}, \quad (26)$$

$$\text{F1} = \frac{2 \text{ Precision Recall}}{\text{Precision} + \text{Recall}}. \quad (27)$$

If a checker sees 100 examples with $TP = 36$, $TN = 54$, $FP = 6$, and $FN = 4$, then accuracy is 0.90, precision is $36/42 \approx 0.857$, recall is $36/40 = 0.90$, and F1 is approximately 0.878. AUROC is the area under the curve traced by true-positive rate against false-positive rate as the decision threshold changes (Fawcett, 2006).

For the multiclass Gate 1 experiment, precision, recall, and F1 are aggregated over error classes by the experiment code. A publication should state whether an average is macro, micro, or weighted whenever the raw report is used. The distilled dashboard values should not be reverse-engineered into class counts because those counts are not stored in the production summary.

C Threshold and Rubric Registry

Rule	Current value	Reason and limitation
Gate 1 word window	70–420	Rejects very thin or unfocused English explanations; not a multilingual reading standard.
Mean sentence length	≤ 27 words	Flags dense prose; does not compute a validated readability grade.
Gate 1 accuracy	$= 4/4$	Treats factual correctness as noncompensatory; still depends on the review model.
Other Gate 1 dimensions	$\geq 3/4$	Requires acceptable completeness, flow, grade fit, and clarity.
Maximum repairs	1	Bounds cost and prevents open-ended self-revision.
Technical frame threshold	0.42	Screens severe render defects before semantic review.
Gate 2 combined threshold	0.65	Engineering acceptance boundary after required component rules.
Gate 2 weights	0.35 / 0.65	Gives semantic meaning more weight than pixel quality; not learned from outcomes.
Minimum scene duration	3.2 s	Provides a reading and transition floor; accessibility testing is still needed.
Narration pace estimate	2.35 words/s	Fallback timing heuristic used only when measured audio is absent.

Table 4: Current thresholds reproduced from the implementation.

D Detailed Deployment Checklist

1. Fork or obtain administrator access to the GitHub repository.
 2. Confirm that `streamlit_app.py`, `requirements.txt`, and `.python-version` are present.
 3. Run the deployment test command from the test protocol in the main text on Python 3.12.
 4. Open <https://share.streamlit.io/> and authenticate with the repository-owning GitHub account.
 5. Choose repository `allanzhang721/AiLessonStudio`.
 6. Choose branch `agent/streamlit-production`.
 7. Set entrypoint `streamlit_app.py` and Python 3.12.
 8. Do not add a shared provider key unless the deployment owner intends to pay for visitor requests. The current interface accepts one visitor-owned key.
 9. Deploy, inspect build logs, open the public URL, and confirm visibility settings permit the intended audience.
 10. Test both providers: OpenAI should expose text, research, image model, and voice controls; DeepSeek should expose text controls and disable video.
 11. Generate a text-only lesson, an illustrated lesson, a failed-key request, a Gate 1 repair, and a visual regeneration.
 12. Record the Git commit, date, provider models, and any calibrated thresholds with published results.
- Official Streamlit guidance recommends declaring dependencies, selecting the intended Python version, and keeping secrets outside the repository (Streamlit, 2026a,b). Repository credentials are required to

configure the hosted app; a source-code commit alone cannot change an account’s cloud visibility.

E Paper Reproduction and Version Statement

Compile the paper from the repository root with:

```
cd docs/whitepaper
pdflatex visual_lesson_ai_paper.tex
bibtex visual_lesson_ai_paper
pdflatex visual_lesson_ai_paper.tex
pdflatex visual_lesson_ai_paper.tex
```

The compilation expects the PNG experiment figures in `../..paper_figures/`. All architecture and process diagrams are embedded as TikZ source in this file. The output should be archived together with the source, bibliography, figure files, software commit, and a short change log.

This white paper describes the code state in August 2026. Generative AI assisted with organization, editing, and diagram planning. The authors remain responsible for checking the implementation claims, citations, equations, and publication permissions. The uploaded SLaTR manuscript was consulted only for general writing habits and exposition structure. It is marked as an anonymous submission whose sharing is prohibited, so it is neither cited as technical evidence nor reproduced.

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